# United International University (UIU)

## Dept. of Computer Science & Engineering (CSE)

## Final-Term Exam::Trimester::Summer 2022

**Course Code: CSE 425 Course Title: Microprocessor, Microcontroller and Interfacing**

**Sec: (A, B) Total Marks: 40 Duration: 2 hour**

**Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.**

**Question 1: Answer all the questions. (14 Marks)**

a. For microprocessor 80286, the value of the DS register is 30H. The offset value is 21H. G bit is set to 1. Now consider the following table and answer the questions: [3 + 2 + 3]

| Global descriptor table | |
|---|---|
| Index | Address |
| 5H | Base: A07000H limit: 0FFFH access right: 1H |
| 6H | Base: A06000H limit: 0FFFH access right: 0H |
| 7H | Base: A05000H limit: 0FFFH access right: 2H |
| 8H | Base: A04000H limit: 0FFFH access right: 1H |
| 9H | Base: A03000H limit: 0FFFH access right: 0H |
| AH | Base: A02000H limit: 0FFFH access right: 2H |

| Local descriptor table | |
|---|---|
| Index | Address |
| 5H | Base: B70000H limit: 0FFFH access right: 1H |
| 6H | Base: B60000H limit: 0FFFH access right: 0H |
| 7H | Base: B50000H limit: 0FFFH access right: 2H |
| 8H | Base: B40000H limit: 0FFFH access right: 2H |
| 9H | Base: B30000H limit: 0FFFH access right: 1H |
| AH | Base: B20000H limit: 0FFFH access right: 0H |

I. Which entry, table and requested privilege level are selected?
II. **Calculate** the starting and ending physical address of the segment.
III. **Calculate** the physical address of the offset value.

b. **Explain briefly** how virtual memory solves the limited physical memory problem. [2]

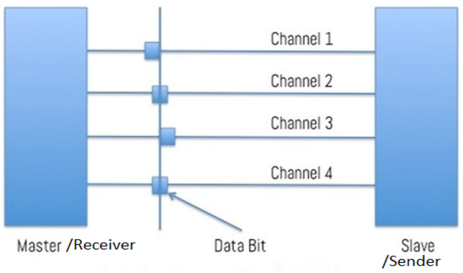
c. Consider the following fetch cycle in an 8086 BIU: [4]

[Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch]

When the first instruction is being executed, two instructions are fetched and saved in the instruction queue. If the 2nd instruction is a **'MOV <address>'** instruction and the 5th is a **'JUMP <7th instruction address>'**, then ***draw*** the corresponding BIU and EU's cycle.

**Question 2: Answer all the questions. (8 Marks)**

a. [3]




**Describe** the phenomenon in the above picture. Explain briefly why it occurs.

b. Master (index:3H) receives 4 byte data (char 'FaiL') from the slave (index:9H). **Draw** the corresponding sequence diagram. [5]

**Question 3: Answer all the questions.** **(10 Marks)**

a. An SAP has following instruction sets: [6]

| Instruction Syntax | Op-code of the Instructions |
|---|---|
| LDA <address> | 1H |
| ADD <address> | 3H |
| SUB <address> | 5H |
| LDB <address> | 7H |
| OUT | 9H |
| HLT | 8H |

Table i : Opcode

| Machine Code | A Register (Initial value 1H) | B Register (Initial value 6H) |
|---|---|---|
| 75H | | |
| 59H | | |
| 17H | | |
| 36H | | |
| 92H | | |
| 81H | | |

Table ii : Machine code

Now, **Determine** the A and B register value after execution of each instruction in table ii

b. Write an assembly code and corresponding machine code for evaluating the following expression. Use the opcode given in **Question 3.a :** 4

-20 - 12 - 5.

**Question 4: Answer all the questions.** **(8 Marks)**

a. An electrical sensor has the four inputs A, B, X and Y where the value of these inputs can be either 0 or 5V. The output of the sensor depends on these four inputs where the output voltage relation is $Vout = A + \frac{1}{2}B + \frac{1}{4}X + \frac{1}{8}Y$. Now, answer the followings: [2 + 2]

i) **Explain in detail** if the above sensor is an Analog or a digital sensor.

ii) If $Vout = 1.25V$ , then **determine** this sensor's input reading.

b. An ADC on an Arduino system converts an input analog signal to the output digital signal as shown in **Fig. 4(c)**. The quantization levels are shown in the figure. [2+2]

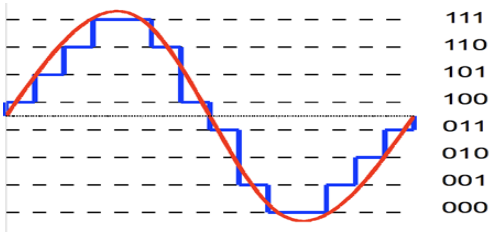



**Fig. 4(c)**

Now, answer the following questions:

i) If the analog voltage is 2.7V, then **determine** the converted digital binary output for a 3.5V Arduino system.

ii) In **Fig. 4(c)**, we can see that the digital signal is not as accurate as the analog signal, **explain briefly** what you should do to get a more accurate digital signal, similar to the analog signal